i-Grad-CAM: Transformation Invariant Grad-CAM

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Introduction

Background

Grad-CAM is a popular interpretability tool for Convolutional Neural Networks (CNNs), but it lacks **robustness** to transformations like rotation, shift, and zoom, leading to **inconsistent explanations [1].**

Grad-CAM and its variants, such as Grad-CAM++ [2] and Score-CAM [3], improved localization and gradient independence but remain **sensitive to basic transformations [4]**. Existing transformation-invariant methods often require **architectural modifications** or extensive data augmentation [5].

Existing methods for transformation invariance are limited because they require **modifications to the CNN architecture** or depend on **additional computational overhead.** A **transformation-invariant approach** is missing, which can ensure **consistent model explanations** without altering the underlying CNN architecture.

Goal

The goal(s) of this work were to:

1. **Develop i-Grad-CAM** to address the sensitivity of Grad-CAM by making its heatmaps **robust** to image transformations, including: **Rotation**, **Shifting**, **Zooming**
2. **Improve interpretability** by ensuring explanations remain **consistent** across transformations.
3. Highlight the utility of the proposed method in **critical applications** like medical imaging, where reliability is essential.

Methodology

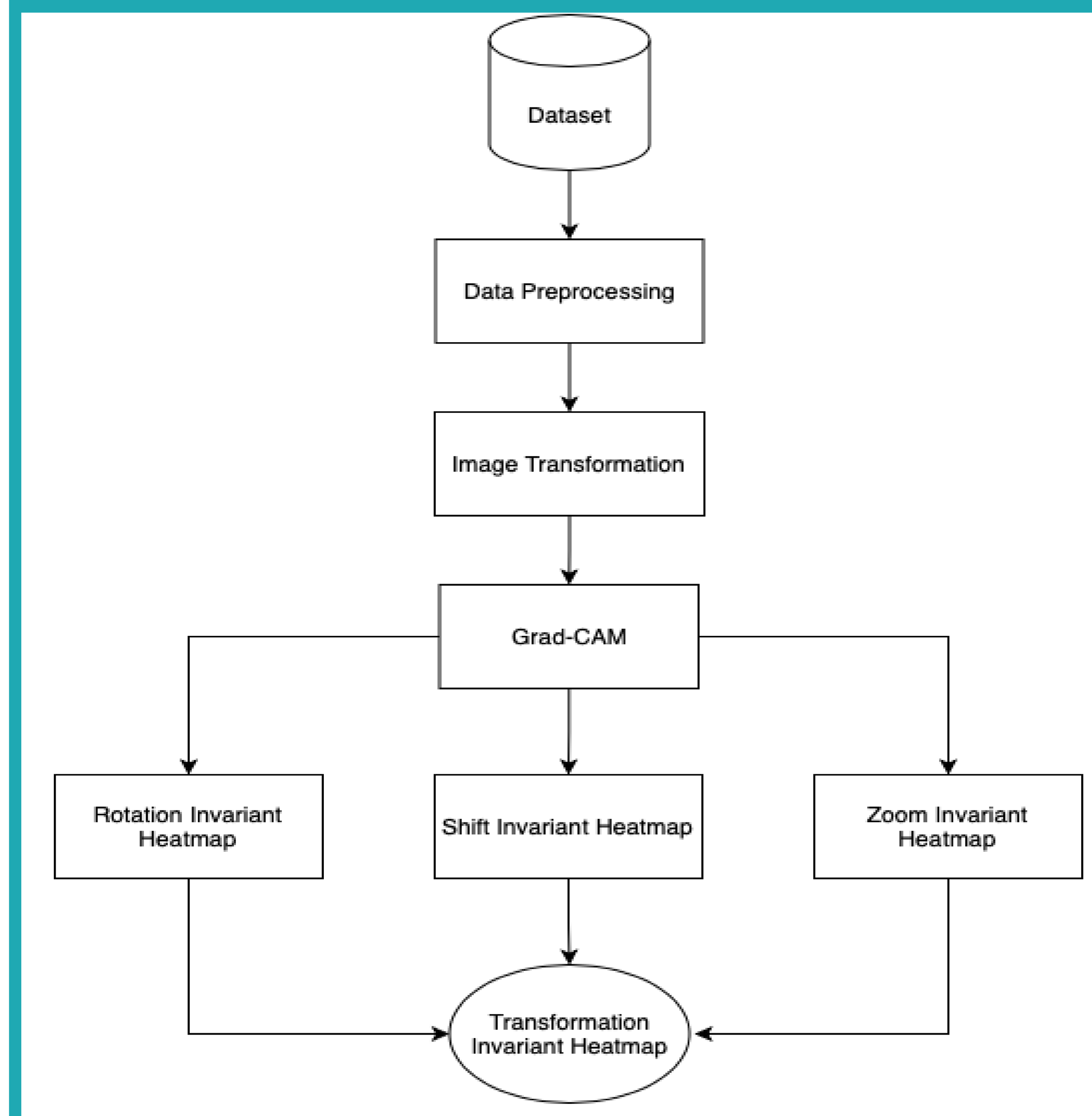

Figure 1. Overview of proposed i-Grad-CAM methodology

Image Transformation and Grad-CAM Generation

For each transformation type (rotation, shift, and zoom), we systematically alter the input image and generate Grad-CAM heatmaps for the modified images.

Rotation Invariance:

i-Grad-CAM achieves rotation invariance by aligning and averaging heatmaps generated from multiple rotated images, ensuring consistent results.

$$H_{avg\_rot} = \frac{1}{No} \sum_{i=1}^{No} H^{i}_{realign}$$

Shift Invariance:

Shift invariance in i-Grad-CAM is achieved by generating and re-aligning heatmaps for systematically shifted images, then averaging them to produce a position-robust heatmap

$$H_{avg\_shift\_y} = \frac{1}{N_y} \sum_{p_y=0}^{H} H^{p_y}_{shift\_realign\_y}$$

$$H_{avg\_shift\_x} = \frac{1}{N_x} \sum_{p_x=0}^{W} H^{p_x}_{shift\_realign\_x}$$

Zoom Invariance:

Zoom invariance in i-Grad-CAM is achieved by averaging Grad-CAM heatmaps generated across multiple zoom levels.

$$H_{avg\_zoom} = \frac{1}{N_z} \sum_{i=1}^{N_z} H^{i}_{zoom}$$

Final Aggregated Heatmap :

The final transformation-invariant heatmap is created by combining rotation, shift, and zoom invariant heatmaps using the Minkowski mean, ensuring robustness to all transformations.

$$\mathrm{H}_{final} = \left(\frac{1}{n} \sum_{i=1}^{n} \|H_i\|^p \right)^{\frac{1}{p}}, \quad \text{where } p = 2$$

where H_i represents each of the four values:

$$H_i \in \{H_{avg\_rot}, H_{avg\_shift\_x}, H_{avg\_shift\_y}, H_{avg\_zoom}\}$$

Results

Figure 1: Traditional Grad-CAM Image

Figure 2: Failure case of Grad-CAM when an image is rotated at 140 degrees

Figure 3. Rotation Invariant i-Grad-CAM Image

Figure 7. Transformation Invariant i-Grad-CAM Image

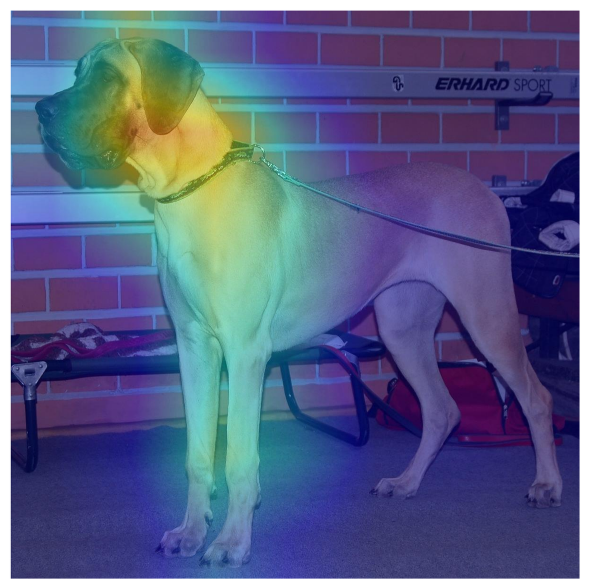

Figure 4. Shift X Invariant i-Grad-CAM Image

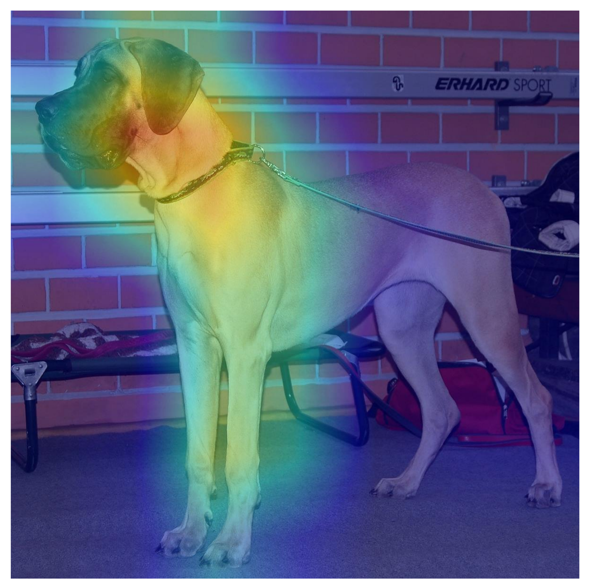

Figure 5. Shift Y Invariant i-Grad-CAM Image

Figure 6. Zoom Invariant i-Grad-CAM Image

Conclusion

We propose i-Grad-CAM, a transformation-invariant extension of Grad-CAM, which improves the consistency and robustness of model explanations across various transformations. By averaging Grad-CAM heatmaps generated from rotated, shifted, and zoomed versions of the input image, our method ensures that the resulting explanation is less sensitive to minor input variations. This approach improves CNN model interpretability by reducing the variability of Grad- CAM heatmaps during image transformations. This transformation-invariant approach is especially useful in fields that rely on precise and consistent model explanations, such as healthcare, where minor differences in input images should not affect interpretative results. Through evaluations, i-Grad-CAM has proven effective in stabilizing heatmaps and producing reliable insights

Future Directions

- Apply i-Grad-CAM for multi-scale abnormality detection in digital pathology.
- Extend i-Grad-CAM to 3D imaging for volumetric tumor analysis.
- Combine i-Grad-CAM with uncertainty maps for improved diagnostic clarity.
- Optimize i-Grad-CAM for real-time clinical use.

References

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